

ADNI Knowledge Graph

Comprehensive Project Report

From Labeled Property Graph to Semantic Knowledge Graph
with Data-Driven Causal Discovery for Alzheimer's Disease

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Metric	Value
Total Nodes	~407,000
Total Relationships	~1,160,000
Unique Patients	2,638
Clinical Visits	30,267
ADNI Tables Integrated	180+
Data Columns	5,800+
Node Types	17
Relationship Types	20+
Image Quality Preservation	100% (MRI & PET)

Dataset: Alzheimer's Disease Neuroimaging Initiative (ADNI) — ADNI-1, ADNI-GO, ADNI-2, ADNI-3, ADNI-4
Technology Stack: Neo4j, Elasticsearch, Redis, Python, Docker
Repository: github.com/theFellandes/ADNIKnowledgeGraph

Table of Contents

1. Executive Summary
2. System Architecture Overview
3. The Core Distinction: LPG vs. Knowledge Graph
4. Patient-Centric Knowledge Graph Schema
5. Node Catalog (17 Node Types)
6. Relationship Catalog (20+ Relationship Types)
7. ADNI Dataset: Table Reference Guide
8. Three-Phase Research Plan
9. Phase 1: In-Place Semantic Migration
10. Phase 2: Causal Discovery Pipeline
11. Phase 3: Validation & AlzKB Integration
12. Ontology Mapping Reference
13. Index & Constraint Strategy
14. Dynamic Graph: Change Detection & Audit Trail
15. Scientific Background & Literature Map
16. Thesis Positioning & Contributions
17. External Resources & Tools

1. Executive Summary

This project constructs a semantically-grounded Knowledge Graph (KG) from the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset, with the ultimate goal of performing data-driven causal discovery on Alzheimer's disease (AD) biomarker cascades. The work operates on a critical distinction that defines its scientific contribution: a **Labeled Property Graph** (LPG) is a data-storage structure optimized for graph queries, whereas a **true Knowledge Graph** carries formal semantic meaning — relationships and nodes are grounded in ontologies (SNOMED-CT, LOINC, UBERON, Relation Ontology), enabling reasoning, inference, and interoperability with external knowledge bases.

The pipeline transforms ADNI's multi-modal clinical, neuroimaging, biomarker, and genetic data into a patient-centric Neo4j KG containing approximately 407,000 nodes and 1.16 million relationships, integrated with Elasticsearch for millisecond-level metadata retrieval and Redis for query caching. The engineering foundation has been validated and published at the **IEEE International Conference on Big Data 2025**, demonstrating 100% pixel-value and bit-depth preservation across all MRI and PET modalities.

The research follows a three-phase plan: **Phase 1 (Construct)** performs in-place semantic migration, adding ontology properties without rebuilding the graph. **Phase 2 (Discover)** applies causal discovery algorithms (PC, FCI, GES, DAG-GNN) to baseline tabular data extracted from the KG. **Phase 3 (Represent)** validates discovered causal edges against AlzKB (118,902 entities, 1.3M relationships) and integrates confirmed edges as first-class CAUSES relationships in the KG. The thesis defense is scheduled for May 2026.

2. System Architecture Overview

The system employs a multi-tier architecture consisting of four integrated components, orchestrated through a 15-step ETL pipeline that processes raw ADNI data — both clinical CSV tables and DICOM neuroimaging files — through sequential stages from database setup through analysis and metrics computation.

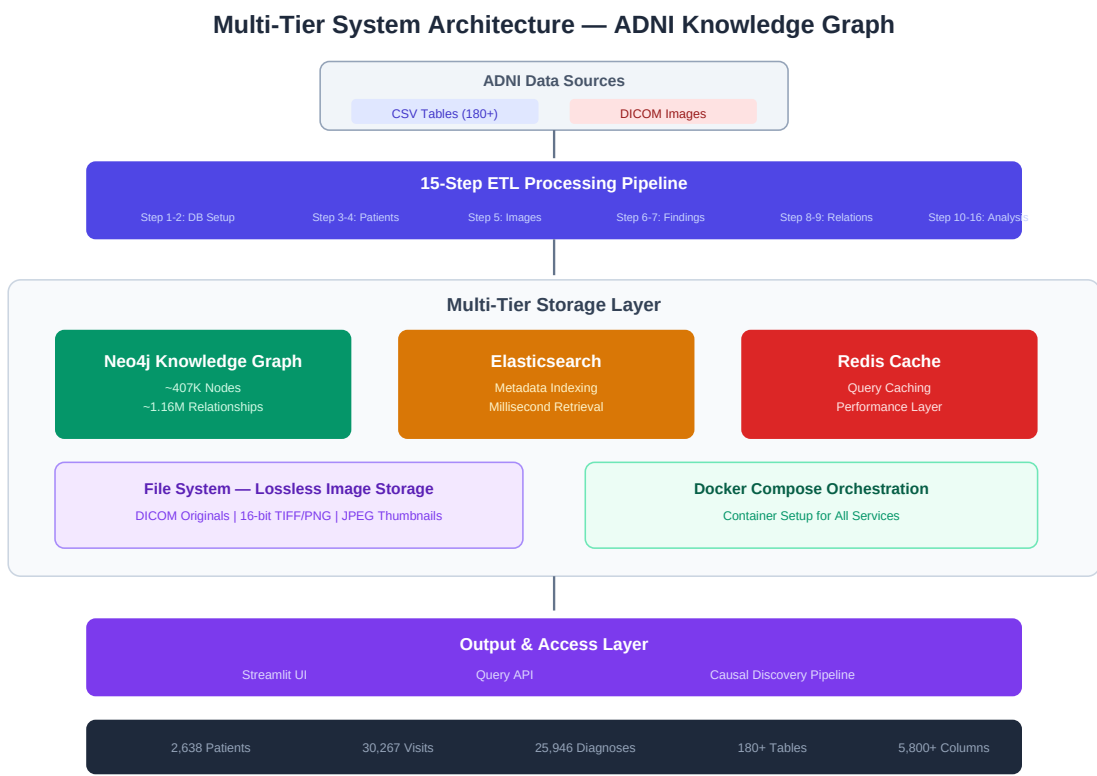


Figure 1: Multi-tier system architecture showing data flow from ADNI sources through the ETL pipeline to storage and access layers.

The **Neo4j Knowledge Graph** organizes patients, visits, diagnoses, images, and biomarkers with ontological annotations, enabling complex cross-domain queries such as retrieving all PET scans with a given biomarker along with subsequent MRI results for the same patient. **Elasticsearch** indexes metadata extracted from DICOM headers and clinical records for millisecond-level query responses. The **multi-tier file system** stores original DICOM images alongside lossless derivatives (16-bit PNG and TIFF) and JPEG thumbnails. **Redis** provides query result caching for frequently accessed patterns.

The 15-step pipeline achieves ingestion rates exceeding 500 images per second while maintaining full data integrity. Each step is independently executable and supports incremental processing via MERGE-based upsert patterns, ensuring the graph can accept new data without requiring a full rebuild.

3. The Core Distinction: LPG vs. Knowledge Graph

Many systems incorrectly claim to be Knowledge Graphs when they are actually Labeled Property Graphs. The key differentiator is **semantic grounding** — whether relationships carry formal, machine-interpretable meaning that enables reasoning and inference. This project's central scientific contribution is the systematic transformation methodology that converts the existing ADNI LPG into a true semantically-grounded KG.

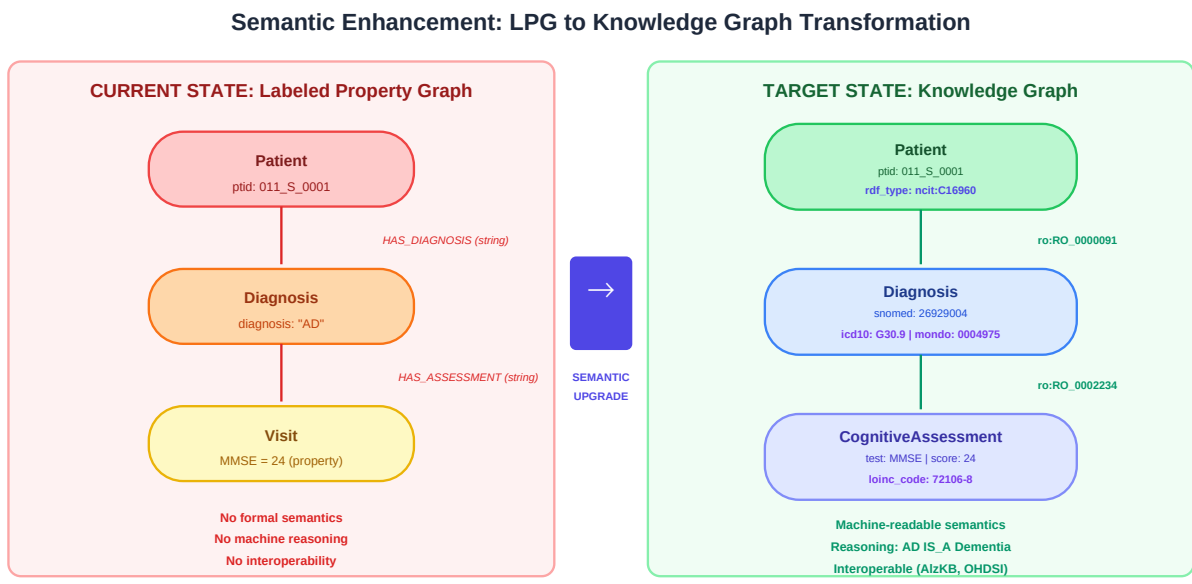


Figure 2: Side-by-side comparison of LPG (current state) and Knowledge Graph (target state) representations.

Aspect	LPG (Current State)	True KG (Target State)
Schema	Implicit, application-defined	Formal OWL/RDF ontology references
Relationships	Typed strings (syntactic)	Ontology-grounded (semantic URIs)
Inference	Not supported	IS_A reasoning, transitive closure
Interoperability	System-specific	SNOMED-CT, LOINC, UBERON, HPO
Disease reasoning	dx_label = "AD" (string)	ICD-10 G30.9 via SPARQL hierarchy
Biomarker semantics	Column names only	LOINC codes as properties
Brain regions	FreeSurfer labels	UBERON codes as properties
Duplicate prevention	Manual checks	Composite unique constraints

Table 1: Detailed comparison of LPG vs. Knowledge Graph characteristics.

The thesis adopts a **hybrid architecture**: most ontological enrichment happens in-place within Neo4j (adding SNOMED, LOINC, UBERON codes as node properties), but disease classification specifically connects to external knowledge bases (ICD-10, MONDO) via RDF/SPARQL. This approach keeps Neo4j's performance for patient-level queries while gaining SPARQL's reasoning capabilities for hierarchical disease classification.

4. Patient-Centric Knowledge Graph Schema

The graph follows a patient-centric design philosophy where individual patients serve as hub nodes connected to their clinical journey: visits, diagnoses, biomarkers, imaging data, medications, adverse events, family history, and genetic profiles. Each patient averages approximately 11.5 visits over the study duration, creating a rich interconnected structure suitable for longitudinal trajectory analysis.

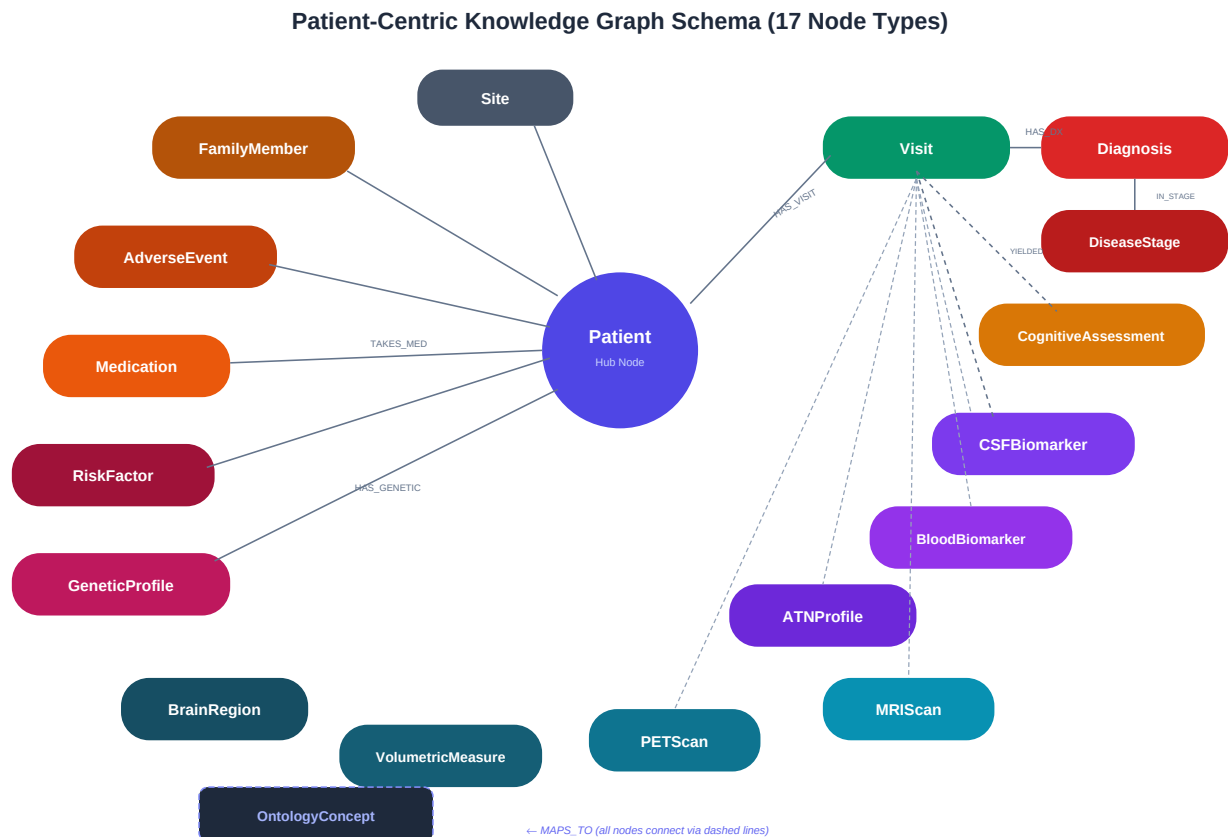


Figure 3: Patient-centric KG schema with 17 node types. Patient is the central hub connected to all clinical domains.

The design supports the complete ADNI data landscape across five major domains: **Clinical** (diagnoses, cognitive assessments, disease staging), **Biomarker** (CSF biomarkers, blood biomarkers, ATN profiles), **Imaging** (MRI scans, PET scans, volumetric measures, brain regions), **Genetic** (APOE genotype, risk SNPs), and **Safety/Treatment** (medications, adverse events, family history, neuropathology). The **Ontology Reference Layer** provides the semantic foundation with OntologyConcept nodes linked via MAPS_TO relationships and organized into IS_A hierarchies.

5. Node Catalog (17 Node Types)

Each node type maps to specific ADNI tables and carries ontology references for semantic grounding. All observation nodes have composite unique constraints to prevent duplicates.

Node Label	Key Properties	Ontology	Source Tables	Constraints
Patient	ptid (PK), rid, sex, birth_year, education	ncit:C16960	PTDEMOG, APOERES	Unique: ptid Idx: rid
Visit	visit_id (PK), viscode, visit_date	ncit:C159705	All tables (join key)	Unique: visit_id
Diagnosis	dx_label, lcd10, snomed, mondo	SNOMED-CT ICD-10	DXSUM, BLCHANGE	Unique: (visit_id, dx_label)
CognitiveAssessment	test_name, total_score, loinc_code	LOINC: 72106-8	MMSE, CDR, ADAS (14 tables)	Unique: (visit_id, test_name)
CSFBiomarker	abeta42, abeta40, tau, ptau, ratio	LOINC: 72333-6	UPENNBBIOMK, BIOMARK	Unique: (visit_id, assay)
BloodBiomarker	analyte, value, units, assay	LOINC per analyte	LABDATA, URMCLABDATA	Unique: (visit_id, analyte, assay)
ATNProfile	a_status, t_status, n_status, atn_class	NIA-AA Framework	Derived from CSF+PET+MRI	Unique: (visit_id)
MRIScan	image_id (PK), sequence, scanner	DICOM + SNOMED	Key_MRI, MRI3META	Unique: image_id
PETScan	image_id (PK), tracer, centiloids	SNOMED: 82918005	Key_PET, AMYMETA	Unique: image_id
VolumetricMeasure	region_name, volume_mm3, thickness	UBERON codes	UCSFFSX7, UCBERKELEY	Unique: (visit_id, region, hemi)
BrainRegion	name, hemisphere, uberon_code	UBERON: 0002421	Reference data	Unique: (name, hemisphere)
GeneticProfile	apoe_genotype, risk_snps	Gene Ontology	APOERES, GENETIC	Idx: apoe_genotype
Medication	compound_name, drug_class, rxnorm	RxNorm / ATC	BACKMEDS, RECCMEDS	Idx: compound_name
AdverseEvent	description, severity, onset_date, outcome	MedDRA codes	ADVERSE, ADSXLIST	Idx: severity
FamilyMember	relationship, has_dementia, age	HPO: HP:0002354	FAMHXPAR, FAMHXSIB	Idx: relationship
NeuropathFinding	braak_stage (0-6), cerad_score, thal	SNOMED neuropath	NEUROPATH (164 cols)	Idx: braak_stage
OntologyConcept	uri (PK), code, label, source	Self-ref (IS_A)	Imported ontologies	Unique: uri Idx: code

Table 2: Complete node catalog with properties, ontology mappings, source tables, and uniqueness constraints.

6. Relationship Catalog (20+ Relationship Types)

Relationship	Source → Target	Properties	Ontology URI
HAS_VISIT	Patient → Visit	—	ro:RO_0000056 (participates_in)
FOLLOWED_BY	Visit → Visit	delta_months	time:intervalBefore
HAS_DIAGNOSIS	Visit → Diagnosis	source_table	ro:RO_0000091 (has_disposition)
PROGRESSED_TO	Diagnosis → Diagnosis	months_between	ro:RO_0002263 (inverse)
TRIGGERED_CONVERSION	Diagnosis → ConversionEvent	—	Custom
YIELDED_ASSESSMENT	Visit → CognitiveAssessment	—	ro:RO_0002234 (has_output)
HAS_CSF_BIOMARKER	Visit → CSFBiomarker	—	ro:RO_0002234
HAS_BLOOD_BIOMARKER	Visit → BloodBiomarker	—	ro:RO_0002234
HAS_ATN_PROFILE	Visit → ATNProfile	—	adni:derived_from
HAS_MRI_SCAN	Visit → MRIScan	—	ro:RO_0002234
HAS_PET_SCAN	Visit → PETScan	—	ro:RO_0002234
HAS_VOLUMETRIC	MRIScan → VolumetricMeasure	—	ro:RO_0002234
MEASURES_REGION	VolumetricMeasure → BrainRegion	—	bfo:BFO_0000066
HAS_GENETIC_PROFILE	Patient → GeneticProfile	—	ro:RO_0000053
TAKES_MEDICATION	Patient → Medication	start, end, dose	ro:RO_0000056
MAPS_TO	Any → OntologyConcept	mapping_confidence	skos:exactMatch
IS_A	OntologyConcept → Onto...	—	rdfs:subClassOf
CLASSIFIED_AS	Diagnosis → ICD-10 (SPARQL)	icd10_code	skos:exactMatch
CAUSES (Phase 2)	Any → Any	algorithm, p_value, effect_size	ro:RO_0002411 (causally_upstream_of)

Table 3: Complete relationship catalog with ontology URI mappings.

7. ADNI Dataset: Table Reference Guide

The headers.json file documents 180+ ADNI tables across 5,800+ columns. Below is a curated reference of the most important tables for KG construction and causal analysis. ADNI data is accessed through the LONI Image and Data Archive (IDA) at ida.loni.usc.edu. Data categories include Subject Characteristics, Clinical Assessments, MRI, PET, Biofluid Biomarkers, Genetics/Omics, and Neuropathology.

7.1 Core Biomarker Tables (Phase 2 Causal Analysis)

Table	Key Columns	Purpose
UPENNBIOMK_ROCHE_ELECSYS	ABETA42, ABETA40, TAU, PTAU	Core CSF biomarkers (Jack cascade pillars)
UCSFFSX7	ST29SV, ST30SV, ST42SV (130+ cols)	FreeSurfer regional brain volumes
UCBERKELEY_AMY_6MM	SUMMARYSUVR_WHOLECEREBNORM	Amyloid PET quantitative SUVR
ADAS	TOTSCORE, TOTAL13	ADAS-Cog (primary cognitive marker)
MMSE	MMSCORE	Mini-Mental State Examination
CDR	CDGLOBAL, CDRSB	Clinical Dementia Rating
MOCA	MOCA (score)	Montreal Cognitive Assessment
APOERES	APGEN1, APGEN2	APOE genotype (critical confounder)
PTDEMOG	PTGENDER, PTDOBY, PTEDUC	Demographics (confounders)
DXSUM_PDXCONV	DXCHANGE, DXCURREN	Diagnosis at each visit

Table 4: Key ADNI tables for biomarker cascade causal analysis.

7.2 Visit Structure

Every ADNI table uses VISCODE (original, e.g., 'bl', 'm06', 'm12') and VISCODE2 (standardized across phases) as visit identifiers. For Phase 2 baseline causal analysis, always filter to **VISCODE2 = 'bl'** to obtain cross-sectional data that avoids longitudinal dependencies. The ABETA42/ABETA40 ratio should be computed as a derived feature before running causal algorithms, as it is a more sensitive amyloid marker than ABETA42 alone.

8. Three-Phase Research Plan

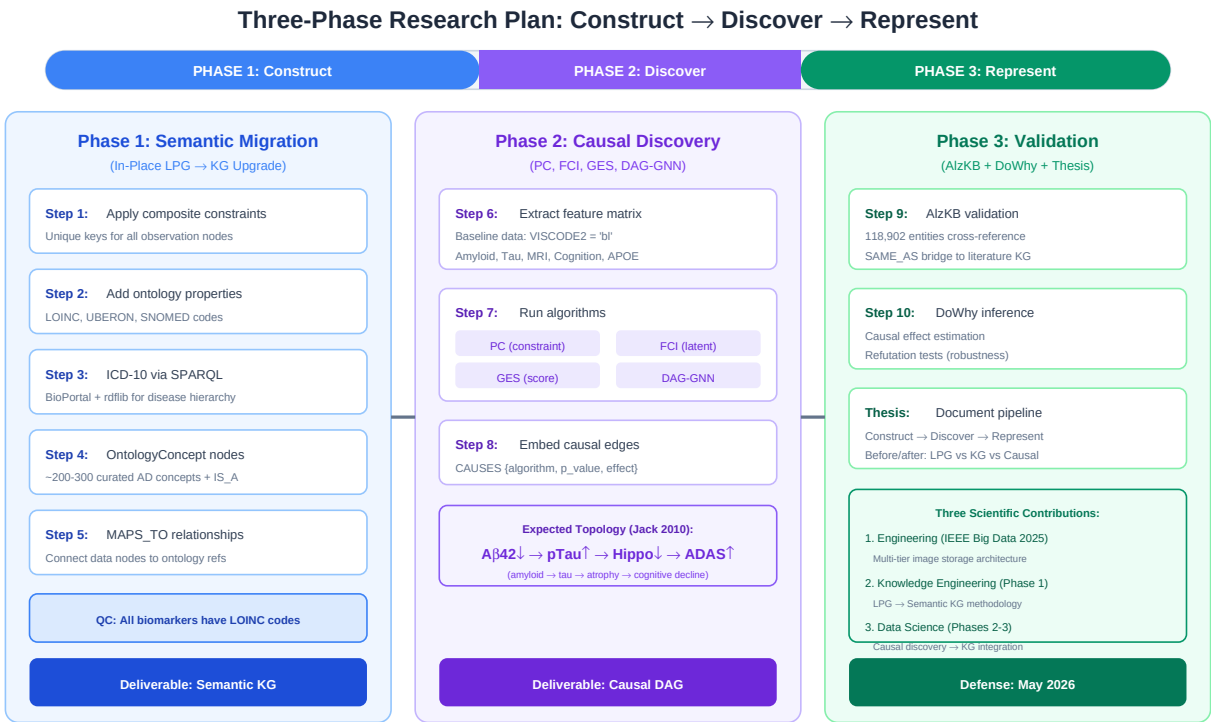


Figure 4: Three-phase research plan — Construct (semantic migration), Discover (causal algorithms), Represent (validation & integration).

The thesis follows a systematic **Construct** → **Discover** → **Represent** methodology, validated by Dobрева et al. (2025) in their systematic review of 43 AD knowledge graph papers. The originality lies in the combination: no existing work integrates a full-scale ADNI data graph (180+ tables, 407K nodes) with semantic ontology grounding AND causal discovery where discovered edges become native KG relationships.

9. Phase 1: In-Place Semantic Migration

The semantic migration transforms the existing LPG into a true Knowledge Graph by adding ontology properties to all node types without rebuilding the database. This in-place upgrade approach preserves the existing 407K-node graph while systematically adding semantic meaning.

9.1 Implementation Steps

Step	Action	Details
1	Apply composite unique constraints	Prevents duplicates: (visit_id, test_name), (visit_id, assay), etc.
2	Add ontology properties (Cypher SET)	loinc_code, uberon_code, snomed_code to all data nodes
3	Resolve ICD-10 via SPARQL	BioPortal endpoint + rdfib for disease hierarchy (AD → G30.9 → G30)
4	Create OntologyConcept nodes	~200-300 curated AD concepts with IS_A hierarchies
5	Create MAPS_TO relationships	Connect biomarker, assessment, brain region nodes to ontology refs

9.2 Ontology Mapping Strategy

Entity Type	Target Ontology	Example Mapping
Diagnosis	SNOMED-CT + ICD-10	AD → SNOMED:26929004 + ICD-10:G30.9
CSF Biomarker	LOINC	Abeta42 → LOINC:57973-5 pTau → LOINC:53817-7
Brain Region	UBERON	Hippocampus → UBERON:0001954
Symptom	HPO	Memory loss → HP:0002354
Medication	RxNorm / ATC	Drug compound → RxNorm code
Relationship	Relation Ontology (RO)	HAS_DIAGNOSIS → RO:0000091

Table 5: Ontology mapping strategy for each entity type.

Quality Criterion: After Phase 1 completion, the query *MATCH (b:Biomarker) WHERE b.loinc_code IS NULL RETURN count(b)* should return 0 for all biomarker nodes with known LOINC mappings. Similarly, all Diagnosis nodes must carry icd10_code, snomed_code, and mondo_code properties.

10. Phase 2: Causal Discovery Pipeline

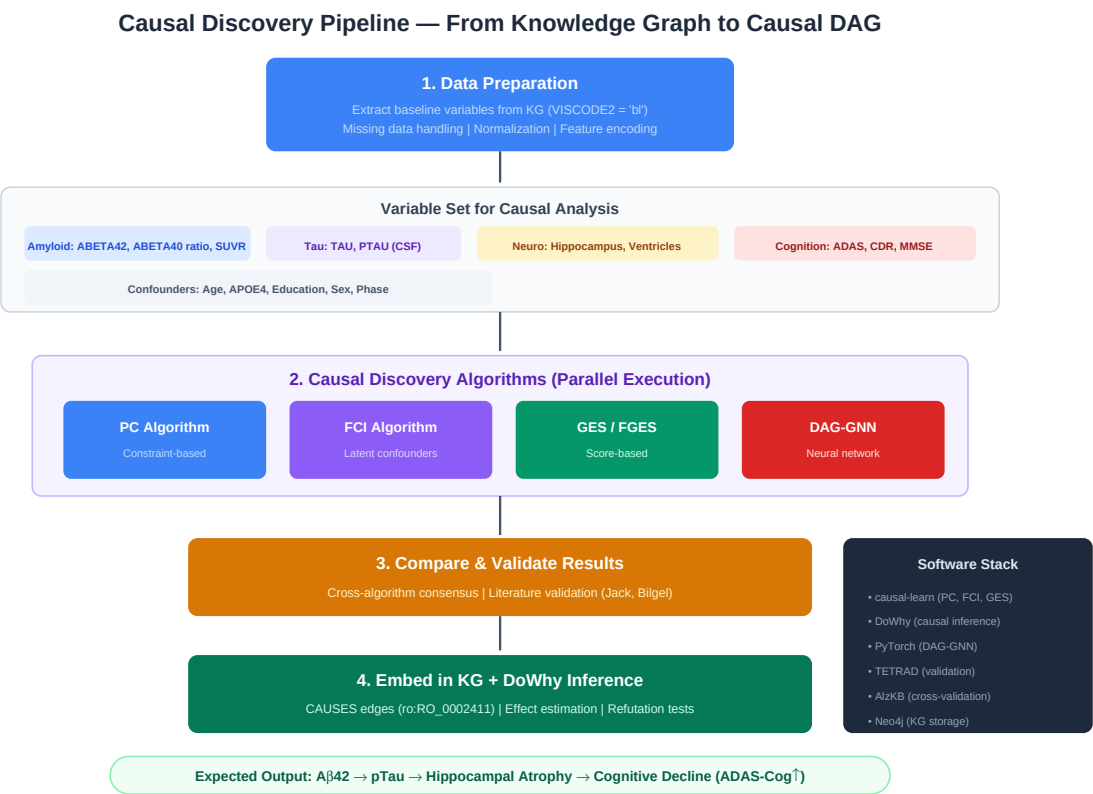


Figure 5: Causal discovery pipeline from data preparation through algorithm execution to KG integration.

10.1 Variable Set

The variable set for causal analysis is carefully selected based on the Jack et al. (2010) biomarker cascade model and validated against Shen et al. (2020) and Bilgel et al. (2022). All variables are extracted from baseline visits (VISCODE2 = 'bl') to ensure cross-sectional independence.

Domain	Variables	ADNI Source
Amyloid (A)	ABETA42, ABETA42/ABETA40 ratio, SUVR	UPENNBBIOMK_ROCHE_ELECSYS, UCBERKELEY_AMY_6MM
Tau (T)	TAU, PTAU	UPENNBBIOMK_ROCHE_ELECSYS
Neurodegeneration (N)	Left/Right hippocampus (ST29SV/ST30SV), Ventricles (ST42SV)	29CSF, 30CSF, 42CSF
Cognition	ADAS-Cog (TOTSCORE), CDR-SB, MMSE	ADAS, CDR, MMSE
Confounders	Age, APOE4 carrier status, Education, Sex	PTDEMOG, APOERES
Cohort	ADNI Phase (1/2/GO/3/4)	ROSTER

Table 6: Variable set for baseline causal analysis.

10.2 Algorithm Comparison Framework

Algorithm	Type	Strengths	Limitations	Software
PC	Constraint-based	Fast, well-understood	Assumes no latent confounders	causal-learn
FCI	Constraint-based	Handles latent confounders	Conservative (partial DAG)	causal-learn

Algorithm	Type	Strengths	Limitations	Software
GES / FGES	Score-based	Scalable, consistent for large feature sets	May overfit with small n	causal-learn / TETRAD
DAG-GNN	Neural network	Captures nonlinear relationships	Requires large data, less interpretable	PyTorch

Table 7: Comparison of causal discovery algorithms.

Expected Topology (based on Jack 2010 + Bilgel 2022): Abeta42 decrease → pTau increase → hippocampal atrophy → ADAS-Cog increase (cognitive decline). FCI is prioritized as the primary algorithm because medical data inherently contains unmeasured confounders (e.g., inflammation, vascular health), and FCI's ability to detect latent common causes makes it the methodologically appropriate choice per Shen et al. (2020).

11. Phase 3: Validation & AlzKB Integration

AlzKB Integration Strategy — Bridging Patient Data and Literature Knowledge



Figure 6: AlzKB integration strategy — bridging patient-level ADNI data with literature-derived knowledge.

AlzKB (Romano et al., 2024) is the largest AD-specific knowledge graph, containing 118,902 entities and 1,309,527 relationships from 21 external databases including DisGeNET, DrugBank, GWAS Catalog, Gene Ontology, and the Monarch Initiative. Our integration creates a bridge between what is known about AD (AlzKB) and what is observed in patients (our KG).

11.1 Validation Protocol

For each causal edge discovered in Phase 2, we perform three validation checks: (1) **AlzKB cross-reference** — query AlzKB for entity pairs matching discovered edges using SAME_AS links between our OntologyConcept nodes and AlzKB entities; (2) **Literature validation** — compare against established findings from Bilgel et al. (2022), Jack et al. (2010), and Shen et al. (2020); (3) **DoWhy refutation** — use the DoWhy framework to estimate causal effects and run robustness checks (random common cause, placebo treatment, data subset tests).

11.2 KG Integration

Validated causal relationships are embedded as first-class **CAUSES** edges in the Neo4j graph with full provenance metadata: algorithm used, p-value, effect size, confidence interval, and validation source. The ontology URI ro:RO_0002411 ('causally upstream of') is assigned to all confirmed causal relationships, enabling semantic reasoning about causal chains across the graph.

12. Ontology Mapping Quick Reference

Entity	Ontology	Code	Label
Patient	NCI Thesaurus	ncit:C16960	Person
Alzheimer Disease	SNOMED-CT	26929004	Alzheimer disease
MCI	SNOMED-CT	386806002	Mild cognitive disorder
CSF Abeta42	LOINC	57973-5	Amyloid beta 42 in CSF
CSF pTau	LOINC	53817-7	Phosphorylated tau in CSF
CSF Total Tau	LOINC	48543-0	Tau protein in CSF
Plasma Abeta40	LOINC	99521-7	Amyloid beta 40 in plasma
Hippocampus	UBERON	0001954	Hippocampal formation
Entorhinal cortex	UBERON	0002728	Entorhinal cortex
MRI scan	DICOM	modality=MR	Magnetic resonance imaging
PET scan	DICOM	modality=PT	Positron emission tomography
APOE4 genotype	Sequence Ontology	SO:0001023	Allele
HAS_PART	Relation Ontology	RO:0000051	has part
PART_OF	Relation Ontology	RO:0000050	part of
CAUSALLY_UPSTREAM_OF	Relation Ontology	RO:0002404	causally upstream of
CORRELATED_WITH	Relation Ontology	RO:0000059	correlated with condition

Table 8: Complete ontology mapping reference for the ADNI Knowledge Graph.

13. Index & Constraint Strategy

The database employs two layers of data integrity: **uniqueness constraints** that enforce entity identity, and **performance indexes** that accelerate the most common query patterns. The indexing rule of thumb is: index every property used in MATCH/WHERE clauses. The most critical path is Patient → Visit → Diagnosis/Biomarker.

13.1 Uniqueness Constraints

Constraint Name	Node	Properties	Purpose
patient_ptid	Patient	ptid	Core entity identity
visit_id	Visit	visit_id	Unique visit instance
mri_image_id	MRIScan	image_id	Unique image reference
pet_image_id	PETScan	image_id	Unique image reference
brain_region	BrainRegion	(name, hemisphere)	Anatomical uniqueness
ontology_uri	OntologyConcept	uri	Ontology term identity
assess_unique	CognitiveAssessment	(visit_id, test_name)	One test per visit
csf_unique	CSFBiomarker	(visit_id, assay)	One assay per visit
blood_unique	BloodBiomarker	(visit_id, analyte, assay)	Unique lab result
vol_unique	VolumetricMeasure	(visit_id, region, hemi)	Unique regional measure
atn_unique	ATNProfile	(visit_id)	One ATN profile per visit
dx_unique	Diagnosis	(visit_id, dx_label)	One diagnosis type per visit

Table 9: Uniqueness constraints ensuring data integrity.

14. Dynamic Graph: Change Detection & Audit Trail

The graph behaves as a living, dynamic system that accepts new data continuously without rebuilding. This is achieved through three layers: **Layer 1** uses SHA-256 hash-based change detection to compare incoming rows against stored hashes; **Layer 2** uses composite unique constraints as a database-level safety net against duplicates; **Layer 3** maintains a BatchIngestion audit trail that logs every ingestion run with statistics on nodes created, updated, and skipped.

The MERGE-based upsert pattern ensures idempotent inserts: every new CSV row or DICOM file follows the pipeline of Parse → Validate → MERGE Patient → MERGE Visit → Create Observation → Add MAPS_TO links. The graph can be re-run on the same data any number of times, and only genuinely new or changed records produce modifications. Every batch is logged with timestamps and counts.

15. Scientific Background & Literature Map

The project draws on seven thematic areas of published research. Each area contributes specific methodology or validation to the thesis.

Area	Key Papers	Contribution to This Project
Foundational Biomarker Model	Jack et al. (2010) Lancet Neurology	Ground truth for A→T→N cascade. DiseaseStage network implements this.
Causal Modeling of AD Cascade	Bilgel (2022), Shen (2020), Zheng (2022), Glazman (2024)	Validates expected causal topology. Informs algorithm selection (FCI).
Multicausality & Systems Models	Uleman (2021), Iturria-Medina (2017), Polsterl (2023)	Justifies multi-modal KG. Warns about unobserved confounders.
AD Knowledge Graphs	Romano (2024) AlzKB, Spasov (2024), Yang (2025), Lyu (2025)	Phase 3 validation target. Methodological precedent. KG + causal integration.
Causal Discovery Algorithms	Ramsey (2017) FGES, Huang (2020) JCI, Gomez (2025) CMC-TS	Phase 2 software selection. Handling multi-cohort data. Partially overlapping variables.
Disease Trajectory Modeling	Young (2018) SuStaln, Bossa (2023) ODE model	Patient subtype awareness. Quantitative benchmark.
Neuroimaging & Multimodal	Raj (2015) Network Diffusion, Venugopalan (2021)	MRI volumetric interpretation. Multi-modal feature importance.

Table 10: Literature map organized by thematic area.

16. Thesis Positioning & Three Scientific Contributions

The thesis makes three scientifically distinct contributions that together represent a novel integration not found in existing literature:

Contribution 1 — Engineering (Published, IEEE Big Data 2025): Multi-tier medical imaging storage architecture combining DICOM preservation, Elasticsearch metadata indexing, and Neo4j ontology-driven graph integration. Validated on ADNI with 100% quality preservation across all MRI and PET modalities.

Contribution 2 — Knowledge Engineering (Phase 1): Systematic transformation methodology for converting a real-world clinical LPG to a semantically-grounded KG. This methodology can be generalized to other clinical datasets. References Spassov et al. (2024) and Dobрева et al. (2025) as prior art being extended.

Contribution 3 — Data Science (Phases 2-3): Data-driven causal structure discovery from ADNI multi-modal data, with results integrated back into the KG as provenance-annotated causal edges. References Lyu et al. (2025) as the methodological framework being instantiated on the AD domain.

Master's Scope (Current)	Future PhD Scope
Construct → Discover → Represent	Predict → Personalize → Validate
Static KG + causal discovery (PC, FCI, GES)	Dynamic temporal causal graphs
Ontology grounding (SNOMED, LOINC, UBERON)	Biomarker cascade simulation (A→T→N)
Baseline-only causal analysis	Personalized trajectory prediction
Literature-validated causal overlay	Clinical trial emulation & validation

Table 11: Master's thesis scope vs. future PhD-level research.

16.1 Originality Statement

No existing work combines: (1) a full-scale ADNI data graph (180+ tables, 407K nodes) with (2) semantic ontology grounding AND (3) causal discovery integration where discovered causal edges become native KG relationships. Spassov et al. had ontology but limited data; AlzKB has scale but no patient-level data; Shen et al. had causal discovery but no KG. This thesis unifies all three.

17. External Resources & Tools

Resource	URL / Location	Use
ADNI Data Portal (IDA)	ida.loni.usc.edu	Download CSVs, images
ADNI Documentation	adni.loni.usc.edu/help-faqs/	Protocol PDFs, manuals
ADNI Data Dictionary	adni.loni.usc.edu/data-samples/ data-dictionary-search/	Column descriptions
AlzKB	github.com/EpistasisLab/ alzheimers-knowledge-base	Phase 3 validation
causal-learn	causal-learn.readthedocs.io	PC, FCI, GES Python
DoWhy	py-why.github.io/dowhy/	Causal inference + refutations
TETRAD	bd2kccd.github.io/docs/tetrad/	Causal discovery suite
Biolink Model	biolink.github.io/biolink-model/	Biomedical KG schema
OBO Foundry	obofoundry.org	UBERON, RO, SO ontologies
LOINC Browser	loinc.org/search/	LOINC code lookup
SNOMED Browser	browser.ihtsdotools.org	SNOMED-CT lookup
GitHub Repository	github.com/theFellandes/ ADNIKnowledgeGraph	Source code

Table 12: External resources and tools used in the project.

Last updated: February 2026 | This report integrates ADNI official documentation, GitHub repository architecture, headers.json dataset schema (180+ tables, 5,800+ columns), IEEE Big Data 2025 paper findings, Design Blueprint v1 and v2 specifications, and the full paper repository bibliography. Prepared for thesis advisory committee review.